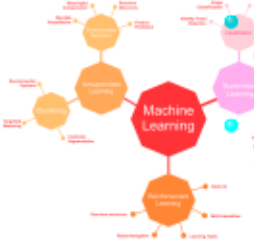
Introduction to Machine Learning

ML-Basics
What is Machine Learning?



Learning goals

- Understand basic terminology of and connections between ML, AI, DL and statistics
- Know the main directions of ML: Supervised, Unsupervised and Reinforcement Learning

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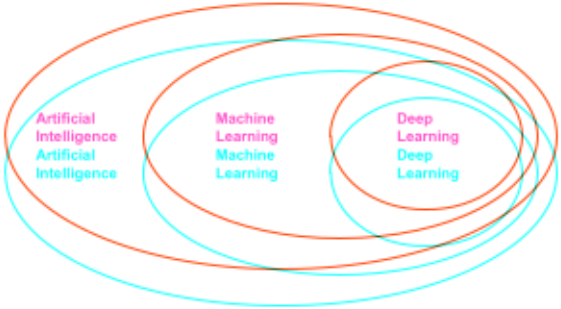
ML IS CHANGING OUR WORLD

- Search engines learn your search preferences
- Recommender systems learn your taste in books, music, movies,...
- Algorithms do automatic stock trading
- Tools can accurately translate between many different languages
- DeepMind beats humans at Go
- Physicians are supported by personalized medicine
- LLMs revolutionize many fields (currently especially coding)
- Data-driven discoveries are made in physics, biology, genetics, astronomy, chemistry, neurology,...
- ...

AI, ML AND DL

THE WORLD OF ARTIFICIAL INTELLIGENCE

... and the connections to Machine Learning and Deep Learning



Many people are confused what these terms actually mean.

And what does all this have to do with statistics?

ARTIFICIAL INTELLIGENCE

- AI is a general term for a very large and rapidly developing field.
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- There is no strict definition of AI, but it's often used when machines are trained to perform on tasks which until that time could only be solved by humans or are very difficult and assumed to require "intelligence".
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- AI started in the 1940s - when the computer was invented. Scientists like Turing and Von Neumann immediately asked: If we can formalize computation, can we use computation to formalize "thinking"?
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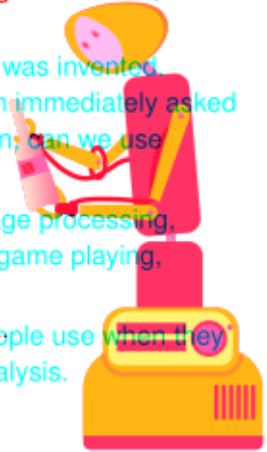
- AI includes machine learning, natural language processing, computer vision, robotics, planning, search, game playing, intelligent agents, and much more.
- Includes ML, NLP, computer vision, robotics, planning, search, intelligent agents, ...
- Nowadays, AI is a "hype" term that many people use when they should probably say: ML or ... basic data analysis.
- Sometimes misused as a "hype" term for ML or basic data analysis.
- Or people refer to the fascinating developments in the area of foundation models



MACHINE LEARNING

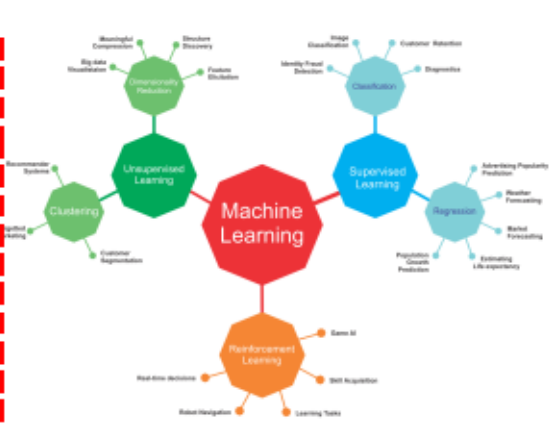


Image via https://www.oreilly.com/library/view/java-deep-learning/9781788997454/assets/899ceaf3-c710-4575-ae99-33c75cd5ac2f.png

- Mathematically well-defined and solves reasonably narrow tasks.
- Usually construct predictive models from data, instead of explicitly programming them.
- "A computer program is said to learn from experience E with respect to some task T and some performance measure P, if its performance on T, as measured by P, improves with experience E."
  Tom Mitchell, Carnegie Mellon University, 1998

DEEP LEARNING

- DL is a subfield of ML which studies neural networks.
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- Artificial neural networks (ANNs) might have been (roughly) inspired by the human brain, but they are simply a certain model class of ML.
- Artificial neural networks are roughly inspired by the human brain, but we treat them as useful, mathematical models.
- ANNs have been studied for decades. DL uses more layers, specific neurons were invented for images and tensors and many computational improvements allow training on large data.
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- DL can be used on tabular data, but typical applications are images, texts or signals.
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- The last 10-15 years have produced remarkable results and imitations of human ability, where the result looked intelligent.
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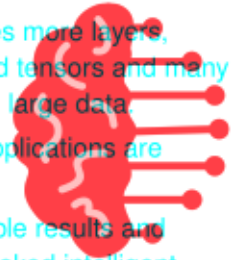

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"Any sufficiently advanced technology is indistinguishable from magic." *Arthur C. Clarke's 3rd law*

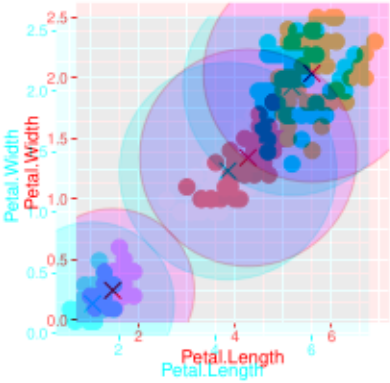
ML VS. STATS

- ML and Statistics have historically been developed in different fields, but many methods and especially the mathematical foundations are equivalent.
- Traditionally, models from ML focused more on precise predictions whereas models from statistics focused more on the ability to derive sound inference.
- Nowadays, ML and predictive modelling in statistics basically work on the same problems with the same tools.
- Unfortunately, the communities are still divided, don't talk to each other as much as they should and everyone is confused due to different terminology for the same concepts.
- Most parts of ML we could also call: Nonparametric statistics plus efficient numerical optimization.

- Historically developed as different fields, but many methods and concepts are pretty much the same.
- ML: Rather accurate predictions with more complex models.
- Stats: More interpreting relationships and sound inference.
- Now: Both basically work on same problems with same tools.
- Communities are still divided.
- Often different terminology for the same concepts.
- Most parts of ML we could also call: Nonparametric statistics plus efficient numerical optimization.
- Personal opinion: Nowadays few practical differences, seeing differences instead of commonalities mainly holds you back

UNSUPERVISED LEARNING

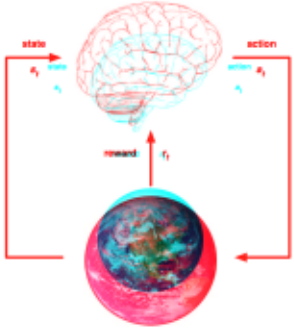
- Data without labels y
- Search for patterns within the inputs $\mathbf{x}$
- *Unsupervised* as there is no "true" output we can optimize against



- Dimensionality reduction (PCA, autoencoders ...); compress information in $\mathcal{X}$
- Clustering: group similar observations
- Outlier detection, anomaly detection
- Association rules

REINFORCEMENT LEARNING

General-purpose framework. At each time step an *agent* interacts with an *environment*. It observes state, receives reward, executes action.



- Goal: Select actions to maximize future reward.
- Reward signals may be sparse, noisy and delayed.

WHAT COMES NEXT

- **Supervised learning** for regression and classification: predict labels y through features $\mathbf{x}$, based on training data.
- First, we will go through fundamental concepts in supervised ML:
  - What kind of "data" do we learn from?
  - What is a "prediction model"?
  - How can we quantify "predictive performance"?
  - What is a "learning algorithm"
  - How can we operationalize learning?
- We will also introduce first concrete learning algorithms: Linear models, trees and forests.
- More complex stuff comes later.

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- We will also look at a couple of fairly simple ML models to obtain a basic understanding.